EVIDENCE · CANVAS · PROSE

math

Boardroom-quality math layouts for mathematicians, quants, ML researchers, physicists, statisticians, and economists. Rendered equations with persona-appropriate surround. Lattice typesets with KaTeX and marp-core with MathJax; the layouts style both, so you author identically either way.

LINEAR REGRESSION · OLS

One equation, displayed; its symbols named below.

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^\top X$ — Gram matrix, $p \times p$, must be invertible

LOGISTIC REGRESSION · MLE

feature crowns the equation full-canvas.

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log\big(1 - \sigma(x_i^\top \beta)\big) \right]$$

WHERE

- ℓ — log-likelihood, concave in β
- σ — logistic link, $\sigma(z) = 1/(1+e^{-z})$
- y_i — observed label, $\in \{0,1\}$
- x_i — feature vector for observation i

derivation walks the steps line by line.

$f(x+h) = f(x) + f'(x)\,h + O(h^2)$	*Taylor expansion,* $n = 2$
$f(x+h) - f(x) = f'(x)\,h + O(h^2)$	*subtract* $f(x)$
$\dfrac{f(x+h) - f(x)}{h} = f'(x) + O(h)$	*divide by* $h \neq 0$
$\lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h} = f'(x)$	**take the limit**

theorem boxes the statement and its proof.

> **DEFINITION.** A function $f : [a, b] \to \mathbb{R}$ is *continuous* on $[a, b]$ if $\lim_{x \to c} f(x) = f(c)$ for every $c \in [a, b]$.

> **THEOREM.** Let f be continuous on $[a, b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ with $f(c) = y$.

> **PROOF.** Set $S = \{x \in [a, b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$. $\square$

compare sets two formulations side by side.

FREQUENTIST

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} p(y \mid \theta)$$

Maximizes the likelihood — no prior. Uncertainty quantified by the sampling distribution of $\hat{\theta}$ across hypothetical repeats.

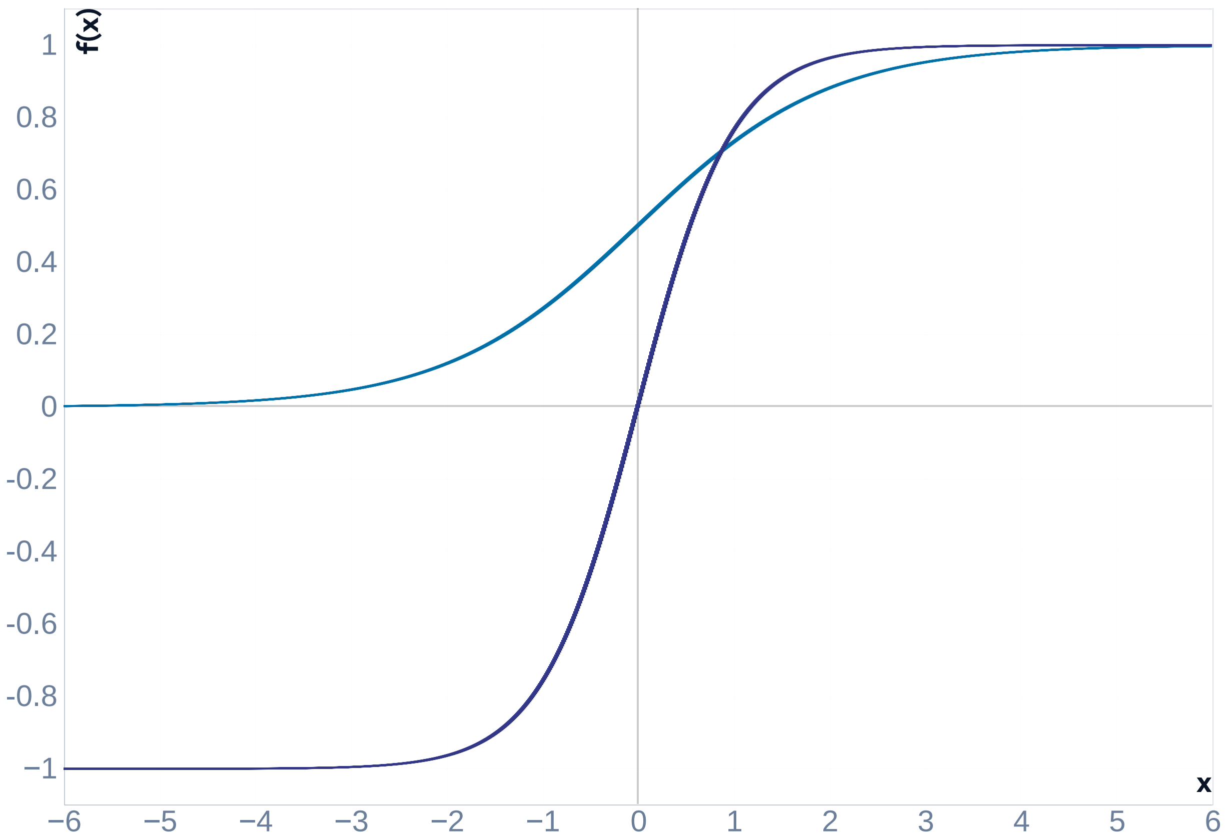
BAYESIAN

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(\theta \mid y)$$

Maximizes the posterior — conditions on the prior $p(\theta)$. Uncertainty is the posterior itself, no repeated sampling required.

canvas gives a long derivation the room.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Maps $\mathbb{R} \to (0,1)$. S-shaped, $\sigma(0) = 0.5$, steepest slope at the origin.

matrix typesets the block structures.

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- SHAPE — $n \times (p+1)$
- ROWS — observations
- COLS — intercept + p features
- RANK — full-rank for OLS to have a unique solution
- COLUMN 0 — all-ones, absorbs the intercept

stats pairs the estimator with its variance.

$$\hat{\beta} = \mathbf{0.42} \pm \mathbf{0.03}$$

95% CI: $[0.36,\ 0.48]$

$p < 0.001 \quad \cdot \quad n = 1{,}204$

For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the p-value just rules out chance.

decompose colors the terms it names.

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

- A — the original matrix being factorized
- L — lower-triangular, unit diagonal
- U — upper-triangular
- USE — solve $Ax = b$ by forward then back substitution

Three displayed steps is the longest chain.

MATH · STRESS

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|_2^2$$

$$X^\top X\hat{\beta} = X^\top y$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

- Step one states the objective
- Step two takes the gradient to zero
- Step three is the ceiling — a fourth step means two slides

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When NOT to reach for math.

Two display equations in the base layout

The bare math layout is built around one hero equation. For side-by-side display, use `math compare`. For a derivation chain, use `math derivation`. Stacking two `$$` blocks in the base layout breaks the visual contract.

Symbols without a legend

An equation with three undefined symbols is a puzzle, not a claim. Either every non-trivial symbol gets a legend entry, or the equation is simple enough that the audience knows it cold.

ASCII math instead of real math markup

Writing `beta_hat = (X'X)^-1 X'y` as plain text bypasses the renderer. Always wrap math in `$$…$$` (display) or `$…$` (inline) — typeset math is the entire reason this layout exists.

RELATED COMPONENTS

See also.

`code` — the implementation, not the equation, is the argument

`diagram` — the structure of the model, not its closed form

`stats` — a row of statistical results without a single equation focus

`content` — one inline equation inside a paragraph of prose